

Shunpeng Bai

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Education

Huazhong Agricultural University

B.S. IN BIOENGINEERING

• GPA: 3.37/4.0

Wuhan, China

September 2021 - June 2026

Experience

Green Pesticide Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. HUAILONG TENG

Apr. 2022 - Oct. 2023

- “Synthesis of remote fluoroalkenyl ketones by photo-induced ring-opening addition of cyclic alkoxy radicals to fluorinated alkenes” D Du, H Peng, L He, S Bai, Z Li, H Teng - *Organic & Biomolecular Chemistry*, 2022
 - **Description:** Synthesized several compounds, and learned many experimental skills, including synthesis, isolation, characterization.

Biocatalysis Laboratory

HZAU, Wuhan, China

DIRECTED BY DR. SHUKE WU

Jun. 2022 - Oct. 2023

- Scientific research and training project was initiated titled “Chemico-enzyme cascade method for green and efficient preparation of Nepetalactone and its analogues”
 - **Description:** Directed evolution of P450-BM3 (CYP102A1) to obtain site selectivity for adipochain oxidation. I learned lots of technologies from DNA to protein, modifying protein, simulation *in silico* and detection of enzyme activity.

Natural Pharmaceutical Chemistry Laboratory

HZAU, Wuhan, China

DIRECTED BY ASSISTANT PROFESSOR SHENGZHEN XU

Oct. 2023 - Present

- “An organic small molecule fluorescent probe for nondestructive detection of Zn²⁺ in plants” J Zheng, S Li, S Bai, X Liu, R Sun, S Xu, W Qu-*Tetrahedron Letters*.2025
 - **Description:** In this project, I undertook the work of theoretical calculation, including conformational search, DFT optimization and single point energy calculation for the probes before and after the combination. The reason why HOMO-LUMO gap becomes larger is explained. I learned a lot of theoretical calculation software and the use of computer cluster, and I wrote some scripts for convenient research.

Shen-Lab

NTU, Singapore

DIRECTED BY ASSISTANT PROFESSOR XIAOTAO SHEN

Mar. 2025 - Aug. 2025

- “Comprehensive Decoding Causal Relationships Between Molecules in Humans Using High-Density Microsampling Multi-Omic Data”
 - **Description:** I am working on this project with Xiaotao Shen.

Awards & Honours

- Summer internship experience at Shen-Lab (NTU) (07.10.2025 - 08.25.2025)
- Remote internship experience at Shen-Lab (NTU) (03.20.2025 - 07.09.2025)
- 2024 Gold Award in the Synthetic Biology Innovation Competition
- Organic lecture titled “Understanding organic reaction from Markov addition reaction mechanism” for undergraduate student
- Third Prize in the 15th “Huazhong Cup” College Students’ Mathematical Modeling competition
- Award for Excellence in the 14th “Huazhong Cup” College Students’ Mathematical Modeling competition
- Served as a peer tutor for the chemistry course at Huazhong Agricultural University (2022.2-2022.7)

Skills

Programming Python, R, Slurm, LaTeX, Bash, Git, Docker, Github Action

Spoken Languages Chinese (Native), English